

Sam Ethan Mathew

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EDUCATION

Carnegie Mellon University
B.S. in Statistics & Machine Learning, Minor in Artificial Intelligence
Coursework: Operating Systems Concepts, Data Structures, Principles of Imperative Computation, Linear Algebra, Probability & Random Processes, Optimization, Deep Learning, Applied Machine Learning, Discrete Mathematics.
Honors & Awards: MIT Hard Mode (1 of 100 global invitees), CMU ProdHacks (1st Place), Jane Street Estimathon (1st Place).

Pittsburgh, PA · May 2028

INTERNSHIP & RESEARCH EXPERIENCE

Robotics Learning Researcher
OpenDroids

- Ported the R2D3 dual-arm mobile manipulator from Gazebo into NVIDIA Isaac Sim 6.0, rebuilding its full URDF/USD model with PhysX drives, so the team could run high-fidelity contact and whole-body experiments in sim before touching hardware.
- Built a Python control SDK for the ported robot, integrating Lula inverse kinematics, motion planning, and real-time collision checking behind one clean interface that runs a complete navigate, grasp, and place mission from a single reusable API call.

San Francisco, CA · May 2026 – Present

Robot Learning Researcher
CMU Robotics Institute, R-PAD Lab (advised by Prof. David Held)

- Lifted tool-use success from 9% to 100% on a real Franka Panda arm, proven over 20 held-out trials on hardware with no sim-to-real transfer, by adapting a learned action-conditioned diffusion world model online from the outcomes it observed.
- Cut the physical interaction needed to learn a new tool tenfold while doubling success, by using epistemic uncertainty from a five-model diffusion ensemble to decide which vision-language model plans were worth executing on the real robot arm.

Pittsburgh, PA · Jan 2026 – Present

Systems & Data Infrastructure Engineering Intern
Solo Tech

- Cut real-time telemetry latency by 35% and log-processing runtime by 60%, by rewriting the services in low-latency C++ and Python and adding automated signal-quality checks that caught corrupt data early, before it could reach the production pipeline.
- Shipped releases 50% more often across a full summer of production deployments, by building Docker containerization and C++ unit-test CI/CD pipelines that caught broken builds automatically and blocked them from merging before review.

Remote · May 2025 – Aug 2025

PROJECTS

Real-Time Voice-Controlled Robot Agent
C++, Python, ROS2, Linux

- Converted a spoken command into physical motion in under 250 milliseconds, timed in live tests on real robot hardware, by building a C++ and Python control pipeline that maps recognized intent onto a structured, validated motion command.
- Ran one shared control interface across two different robot platforms with zero code changes, by validating every command against explicit safety limits and a deterministic fail-safe fallback path before the controller could command any motion.

Aug 2025

Multi-Threaded Concurrent Storage Engine
C++20, Linux, POSIX

- Cut worst-case (P99) write latency by 45% under heavy multithreaded load, by engineering a concurrent LSM-tree storage engine in C++20 with lock-free write buffers, memory-mapped file I/O, and Bloom-filter key lookups on the read path.
- Debugged a write-throughput ceiling nobody could explain, by profiling the live system on Linux with perf and eBPF kernel traces, tracing it to disk-page allocation logic and clearing it with a thread-safe, fully retuned buffer pool manager.

Jan 2026

AI Code-Search Engine (Sweep-AI)
C++, Python, Tree-sitter

- Dropped median code-symbol lookup from 24ms to 7ms while raising recall to 80.6%, by rebuilding a coding assistant's search layer in C++ around a Tree-sitter parser and compact 50-token summaries of every function and type in the repository.
- Shrank the context sent to the model 12×, from 30K to 2.5K tokens, and cut wrong-file retrievals by over 30 percentage points, by adding sub-millisecond intent classification across six task types and a strict, enforced per-query token budget.

Jan 2026

PUBLICATIONS

Submitted to the 10th Conference on Robot Learning (CoRL 2026) — Under review

- Doubled real-world manipulation success while cutting the physical trials required tenfold, by pairing a vision-language planner with a learned action-conditioned world model that predicts each plan's outcome before the arm ever executes it on hardware.
- Designed the dual-role uncertainty strategy at the core of the method, by having a five-model diffusion ensemble both select informative plans for training and gate unreliable plans at execution, adapting the world model from about 20 plans per task.

LEADERSHIP

Head of AI, ScottyLabs (Labrador Incubator)

2025 — Present

Carnegie Mellon University

- Set the technical direction for a student-run software organization's AI arm, running design and code reviews for every team inside its Labrador incubator and translating hard engineering trade-offs into clear terms a non-specialist can act on.

SKILLS

Languages: Python, C++ (C++14/17/20), C, CUDA, Java, SQL, Go.

Robotics & ML: PyTorch, Diffusion Models, World Models, Vision-Language Models, Reinforcement Learning, ROS2, NVIDIA Isaac Sim, Gazebo, URDF/USD, Inverse Kinematics, Motion Planning, State Estimation, Franka Panda.

Systems: Linux, Multithreading, Lock-Free Concurrency, Real-Time IPC, Debugging & Profiling (perf, eBPF), Git, CI/CD, Docker, Unit Testing.

Math: Linear Algebra, Optimization, Probability.